Closed-Loop BCI-Triggered System with Soft-Robotic Glove for after-stroke rehabilitation

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1 Clinical Need

Stroke represents a major cause of long-term neurological disability globally. As evidenced by epidemiological data from the World Stroke Organization [1], the socio-health impact of this pathology is constantly growing: over 93 million cases prevalent, some 160 million years of life lost to disability (DALYs) and a estimated global economic cost of more than US$ 890 billion.

Between 55 % and 75 % of survivors have chronic upper limb motor deficit which compromises autonomy in daily activities [12]. Donkor [2] contextualizes this figure in the perspective of quality of life: stroke is the second cause of death in the world and causes severe permanent disability in about half of the survivors, with high impact on autonomy, working life and the care burden on family members.

The actual time spent on active therapeutic exercise (time on task) within conventional sessions varies on average between 8 and 44 minutes per day [3] — significantly lower than recommended doses to induce stable cortical reorganization.

2 Current Standard of Care and Its Limitations

The current standard of post-stroke upper limb rehabilitation combines conventional physiotherapy with commercially available robotic orthoses. Among passive systems: Bioservo SEM Glove uses EMG-threshold triggering on a peripheral signal absent in dense hemiparesis; Gloreha applies passive mobilization on a fixed schedule; Hand of Hope employs EMG amplitude as a proportional actuator [15]. All three produce joint mobilization without cortical engagement.

Wang et al. [7] quantify this failure: a passive soft-robotic glove improved FMA-UE scores but produced no significant changes in Resting Motor Threshold or Motor Evoked Potential amplitude — the markers of cortical synaptic excitability. The glove moves the hand; it does not engage the brain. After discharge, most patients receive virtually no structured rehabilitation [3].

The mechanism required for genuine recovery is well established: stable synaptic modification (STDP) requires temporal coincidence within $\approx$300 ms between cortical motor intent and afferent feedback [4, 5]. A random or sham activation produces no β-band coherence increase [5] and no clinical improvement [6]. Rustamov et al. [11] confirm that theta-gamma CFC over C3/C4 increases proportionally to motor recovery across BCI sessions — the gap is not biological, it is a delivery problem.

The sole BCI-triggered orthosis to reach clinical deployment, IpsiHand (Neurolutions), closes this neural loop, but was validated exclusively in supervised clinical settings with trained personnel at each session [11]. No commercial system today implements contingent BCI triggering in an unsupervised, home-deployable form.

3 The Human Limitation Being Addressed

The mechanism described above requires three conditions that no human operator can reliably provide in an unsupervised home setting:

(1) Continuous presence. The Hebbian contingency must be enforced at every repetition across sessions of 30–45 minutes, 3–5 times per week, for months. A trained therapist cannot be physically present at the patient's home at this frequency — this is precisely the dose gap documented by Newton et al. [3].

(2) Sub-300 ms reactive latency. The BCI trigger must fire within the Hebbian coincidence window [5]. No human operator can detect ERD onset and manually trigger mechanical feedback within this window reliably across hundreds of repetitions — an automated closed-loop system can satisfy this constraint consistently.

(3) Operator-independent consistency. Outcome variability in conventional rehabilitation is partially attributable to inter-therapist inconsistency in task delivery and feedback timing [3]. A closed-loop BCI system applies identical contingency criteria at every trial, eliminating this source of variance. Ren et al. [9] confirm in a meta-analysis of 10 RCTs (290 patients) that BCI-triggered intervention outperforms conventional therapy (SMD = 0.50, 95 % CI: 0.26–0.73), with effect size directly linked to contingency fidelity.

The robot is not a substitute for clinical judgment — the therapist remains in the loop asynchronously via remote dashboard. It is a substitute for physical presence, consistency, and automatic reactivity.

4 Proposed Robotic Solution

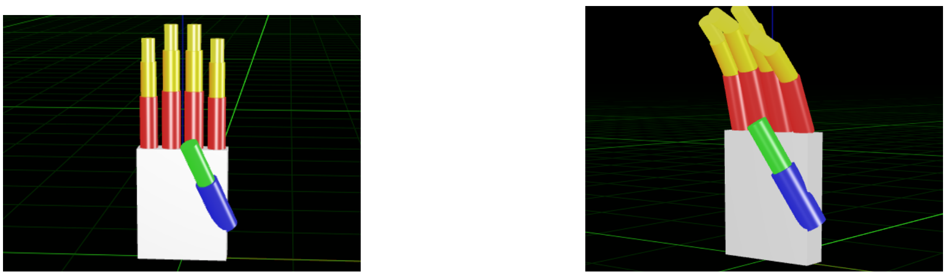
The proposed solution — a BCI-Triggered Soft Robotic Glove (BCI-SRG) — is the direct engineering consequence of the three limitations identified above. The device is a wearable, unsupervised home system used in autonomous sessions of 30–45 minutes, 3–5 times per week, without on-site clinical presence. A GUI manages session setup; the therapist supervises asynchronously via cloud dashboard.

4.1 Kinematic Architecture

The architecture is a soft-robotic pneumatic wearable glove, modelled as a system of serial kinematic chain anchored to a fixed palm [12, 15]. The choice of using a pneumatic network (PneuNet) is dictated by the need of intrinsic security and passive adaptability to variable geometries in an unsupervised environment [12, 8].

The geometric model replicates human anatomy with 3 Degrees of Freedom (DoFs) for each of the 4 fingers (MCP, PIP, and DIP joints) and 2 DoFs for the thumb (CMC and MCP joints), reaching a total of 14 kinematic DoFs. However, the system is intentionally under-actuated, employing a single independent pneumatic channel per digit to yield 5 active DoFs. The complexity-benefit trade-off of this design is governed by three primary factors:

1. **Biological Synergies:** Neurophysiological studies (e.g., Santello et al. [17]) demonstrate that the central nervous system controls hand grasping through low-dimensional synergistic patterns rather than isolated joint control. Actuating 14 joints independently is redundant for post-stroke rehabilitation, which focuses on restoring gross functional synergies for Activities of Daily Living (ADLs).
2. **Embodied Intelligence:** The PneuNet architecture exploits morphological computation. Upon pressurization, fluid redistribution within the elastomeric chambers naturally conforms to the object's geometry, bypassing the need for high-density sensor arrays or complex multi-joint control algorithms.
3. **Usability and Weight Constraints:** Implementing 14 active lines would introduce excessive hardware overhead. Minimizing weight is crucial to prevent muscular fatigue and abnormal compensatory movements at the shoulder, ensuring autonomous, single-handed donning by hemiparetic patients at home.



(a) Frontal view (b) Lateral view, showing joint flexion

Figure 1: URDF kinematic model of the robotic glove.

Mathematically, this under-actuation defines a constrained kinematic mapping where the joint space $q \in \mathbb{R}^{14}$ is coupled to the actuator pressure space $u \in \mathbb{R}^5$ via a non-linear vector function:

$$q = g(u) \tag{1}$$

where the passive elasticity of the silicone elastomer acts as a virtual restoring spring, guaranteeing a unique, statically stable equilibrium configuration for each input pressure vector u.

4.2 Sensing and Biomechanical Safety

System sensing is categorized into two levels: neurological sensing (EEG) and local biomechanical sensing. To ensure patient safety in the absence of a clinician, the firmware implements three quantitative mechanisms in compliance with active medical device standards (IEC 60601-1) and IEC 62304 software guidelines. The system is designed to achieve the Performance Level required by ISO 13849:

- **Admittance Control ($F \to v$):** Integrated palmar piezoelectric sensors measure interaction forces, dynamically modulating pneumatic pressure to ensure glove compliance; the system automatically yields to spastic resistance, prioritizing patient safety and joint protection. Kinematic joint boundaries defined in the URDF model serve as software-enforced virtual fixtures.
- **Fail-Safe EEG:** The system continuously monitors electrode impedance and signal quality (SNR). In the event of signal loss (1 s timeout), all pneumatic valves are exhausted within 200 ms, ensuring the device cannot execute non-intentional autonomous movements.
- **Emergency Stop:** A physical, high-visibility emergency button is located on the device controller. Accessible with the patient's unaffected hand, it immediately cuts power to the valves and exhausts residual pneumatic pressure, independently of the software state.

The neurological sensing layer processes EEG acquired from the rigid-shell headset across four sub-bands (4–8 Hz, 8–12 Hz, 13–30 Hz, 30–45 Hz) using Filter Bank Common Spatial Pattern (FBCSP) [14, 16], producing a compact spatial-spectral feature vector. Classification uses an Adaptive LDA [13] that performs recursive parameter updates online, compensating for intra-session signal drift caused by cortical reorganization, cognitive state fluctuation, and electrode impedance drift, documented sources of EEG non-stationarity in BCI systems for post-stroke rehabilitation [13, 16]. FBCSP feature extraction and LDA classification are computationally tractable for real-time BCI operation [14, 13], and the combined pipeline targets an end-to-end latency well below the 300 ms Hebbian contingency window [5]. Actuation is executed via five solenoid valves (one per digit), with pressure modulated by the admittance controller described above.

4.3 Level of Autonomy, EEG Interface, and Inclusion Criteria

The system operates at **Level of Autonomy 2 (Shared Control)** according to the Troccaz classification: the patient retains high-level decisional control by generating motor intent, while the robot autonomously manages low-level execution (pressure trajectories, force sensor monitoring, and feedback timing).

The system cannot operate at LoA 3 or higher because autonomous actuation, decoupled from cortical intent, would violate the Hebbian plasticity principle. A robotic glove moving without the patient's active intent would provide passive rehabilitation rather than neuroplastic training [4, 5]. Thus, LoA 2 is not a design compromise; it is the autonomy level required to induce neuroplasticity.

To ensure autonomous self-donning, the system utilizes a pre-shaped rigid-shell EEG headset with spring-loaded dry electrodes. The mechanical structure guides the headset into the correct position on the C3/C4 sites, removing the need for conductive gel or high-precision alignment.

Motor imagery cues are triggered synchronously with the patient's inhalation phase via an abdominal strain sensor, stabilizing the EEG baseline and reducing false-positive ERD detections [14].

Inclusion Criteria as Design Specification

Autonomous donning assumes sufficient function in the contralateral hand, which is implicit in the primary clinical inclusion criterion (patients must be capable of single-handed operation). The system is therefore indicated for patients with **unilateral hemiparesis**. Patients with severe bilateral paralysis are excluded by the device's inherent design, not by a technological limitation. This clinical boundary is an intentional design specification.

Remote clinical supervision is conducted via an **asynchronous cloud-based dashboard**, allowing clinicians to regulate virtual fixtures (angular limits, admittance thresholds), review session logs, and personalize task parameters [12].

4.4 Conceptual Diagram

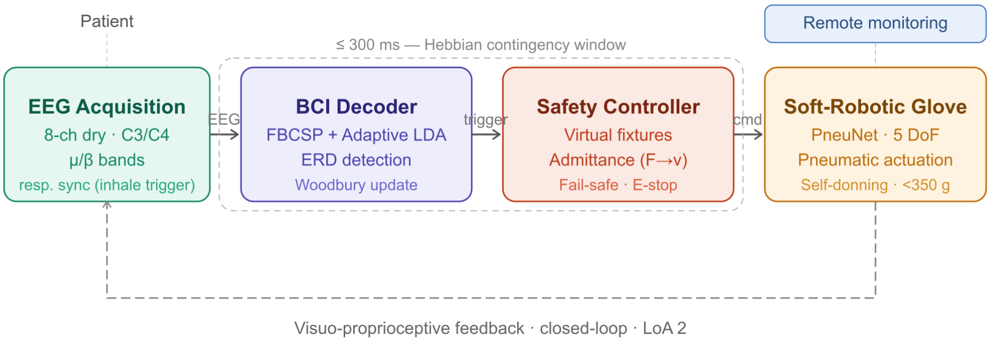


Figure 2: Closed-loop architecture of the BCI-SRG system. Key components: (1) EEG acquisition with respiratory synchronization; (2) FBCSP adaptive LDA decoder; (3) safety controller with virtual fixtures and admittance control; (4) soft pneumatic glove. The dashed arrow represents the visuo-proprioceptive feedback loop. Target latency $\leq 300\,\text{ms}$ [4, 5].

5 Why Feasible Now

Three convergences make this solution feasible now:

- **Dry-electrode EEG at clinical quality:** until 2020, dry electrodes had SNR too low to reliably detect ERD on C3/C4 in real-time. Current systems [14] have filled this gap. Without this, self-donning at home was not possible – a technician was needed for gel.
- **Real-time FBCSP inference on edge hardware:** The FBCSP + LDA pipeline has been demonstrated for real-time BCI control in deployed clinical systems [13, 14]. Five years ago, comparable pipelines required a dedicated workstation with a cabled connection, incompatible with unsupervised home use.
- **Affordable soft robotics:** PneuNet actuators in cast silicone using 3D-printed molds reduce production cost of the glove. Proietti et al. [12] demonstrated in 2024 the first clinical validation of a fully home-based system on 10 patients.

The three enablers above represent mature components ready for integration. What must instead be developed: robust ERD decoding in patients with extensive cortical lesions where Motor Imagery signals are significantly reorganized, and a validated home-safety protocol for the EU MDR Clinical Evaluation Report, which does not yet exist for unsupervised home-use BCI orthoses.

6 Regulatory and Adoption Pathway

The device is classified as **FDA Class II** via the 510(k) clearance pathway, with the IpsiHand (Neurolutions) serving as the predicate device due to its identical BCI-triggered robotic paradigm, as validated in clinical outcomes studies [11].

In the European Union, the device is classified as **Class IIb** under the EU MDR 2017/745 (Rule 10), as it is an active therapeutic device that administers energy to the patient. Compliance requires a comprehensive Clinical Evaluation Report (CER) and assessment by a Notified Body, demonstrating substantial equivalence to current BCI-triggered orthoses.

The value proposition targets rehabilitation centers, not payers directly. In the US, outpatient physical therapy is reimbursed under CPT 97110–97530; the BCI-SRG does not require a new billing code, but substitutes in-person sessions with remotely supervised home sessions, allowing one therapist to monitor multiple patients concurrently and reducing per-patient labor cost without reducing reimbursable therapist oversight.

In Europe, NHS/SSN reimbursement will require inclusion in national device registers (e.g., NICE appraisal, Italian LEA), contingent on RCT evidence — a 3–5 year timeline post-CE mark. A per-session disposable (electrode pad set) provides recurring revenue independent of the reimbursement timeline. The purchasing decision rests with rehabilitation center directors and hospital procurement, not individual clinicians or insurers.

7 Risks and Failure Modes

We identify three substantive risks across clinical, technical, and commercial dimensions, each paired with a specific mitigation strategy:

1. **Technical: Adaptive LDA failure in severe lesions.** The Adaptive LDA compensates for intra-session drift, but patients with extensive M1 infarction may produce ERD signals too weak or spatially reorganized for reliable decoding even after adaptation. In this subpopulation, the mitigation itself may be insufficient. *Mitigation:* Restrict clinical indication to patients with residual corticospinal pathways (confirmed by TMS screening at enrollment); flag sessions with classification confidence below threshold for therapist review via dashboard.

2. **Clinical/Safety: Spasticity severity at enrollment.** The admittance controller mitigates spasticity-induced force during sessions, but patients with severe spasticity (Modified Ashworth Scale, MAS ≥ 3 — a standardized clinical measure of post-stroke muscle tone ranging from 0 to 4) represent an edge case where passive resistance may persistently exceed the therapeutic working range, preventing effective glove actuation altogether. *Mitigation:* Mandatory spasticity screening at enrollment (MAS ≤ 2 as inclusion criterion); automatic session suspension if interaction force exceeds threshold for $>500\,\text{ms}$.

3. **Commercial: Clinician Rejection.** Therapists may perceive BCI-based systems as an unreliable substitute for traditional manual therapy or fear a loss of professional oversight. *Mitigation:* Position the device as an **adjunct tool** for *dose intensification* rather than a replacement. The clinical dashboard provides therapists with granular objective data (ADL success rates, BCI engagement, joint ROM), transforming the BCI-SRG into a digital instrument that empowers, rather than replaces, the physiotherapist's clinical expertise.

8 What This Concept Does Not Solve

Honest limitations of the current BCI-SRG scope:

- **Bilateral paralysis:** The system requires single-handed donning with the contralateral limb; patients with severe bilateral paresis are excluded by design, not by a technological gap.

- **Severe cognitive impairment:** Motor Imagery requires active engagement and metacognitive awareness. Patients with severe post-stroke cognitive decline or aphasia affecting task comprehension are outside the indicated population.

- **Standalone replacement of therapy:** The BCI-SRG is a tool for *dose intensification*, not a standalone cure. It does not replace the complex, multidimensional physical and occupational therapy required for global post-stroke recovery; rather, it operationalizes the high-frequency repetitions that current healthcare systems are unable to provide.

9 Concluding Statement

Stroke leaves millions of patients with a motor deficit that rehabilitation science knows how to address — but that healthcare systems cannot deliver at the required dose, frequency, and temporal precision. The BCI-Soft Robotic Glove does not propose a new treatment: it proposes a delivery mechanism for one that already works, constrained to the exact autonomy level that the underlying neuroscience demands. The concept is deliberately narrow — one disorder, one limb, one mechanism. Its value, if validated, lies not in replacing the therapist but in making the Hebbian contingency window accessible every day, at home, without one.

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